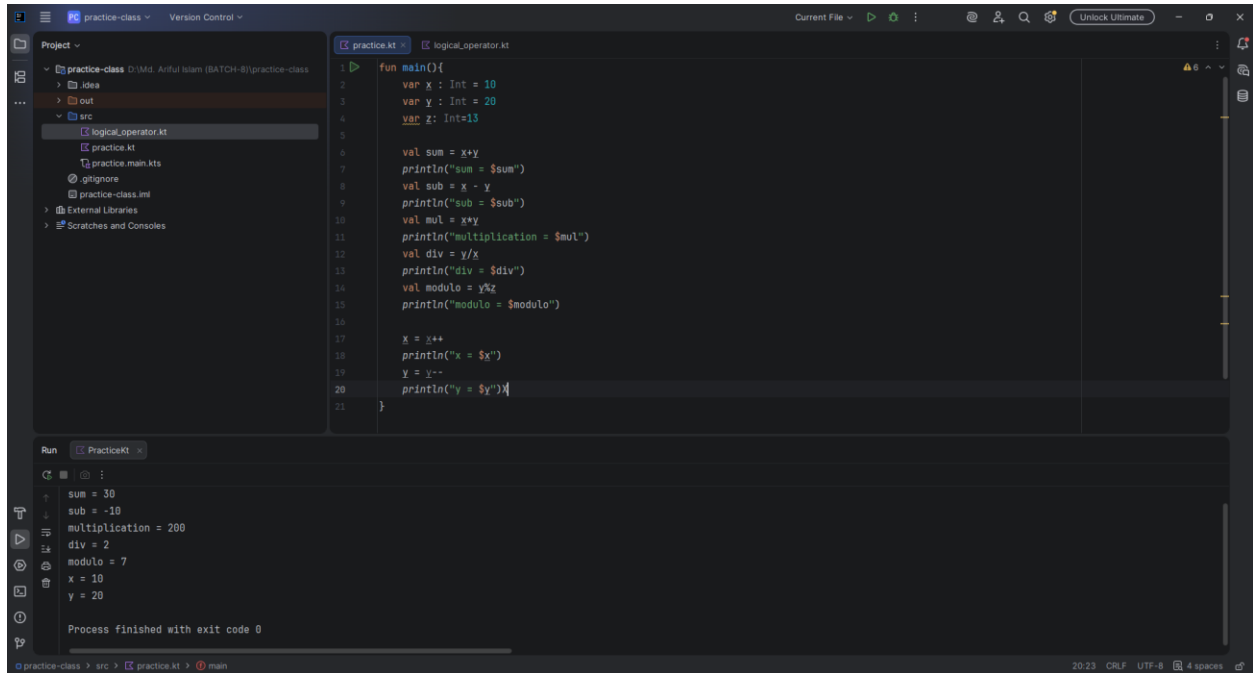


Arithmetic operator



The screenshot shows the IntelliJ IDEA IDE with a project named 'practice-class'. The 'src' directory contains a file 'logical_operator.kt'. The code in this file defines a 'main' function that performs several arithmetic operations on variables 'x', 'y', and 'z'. The operations include addition, subtraction, multiplication, division, and modulo. The results are printed to the console. The Run window at the bottom shows the output of these operations.

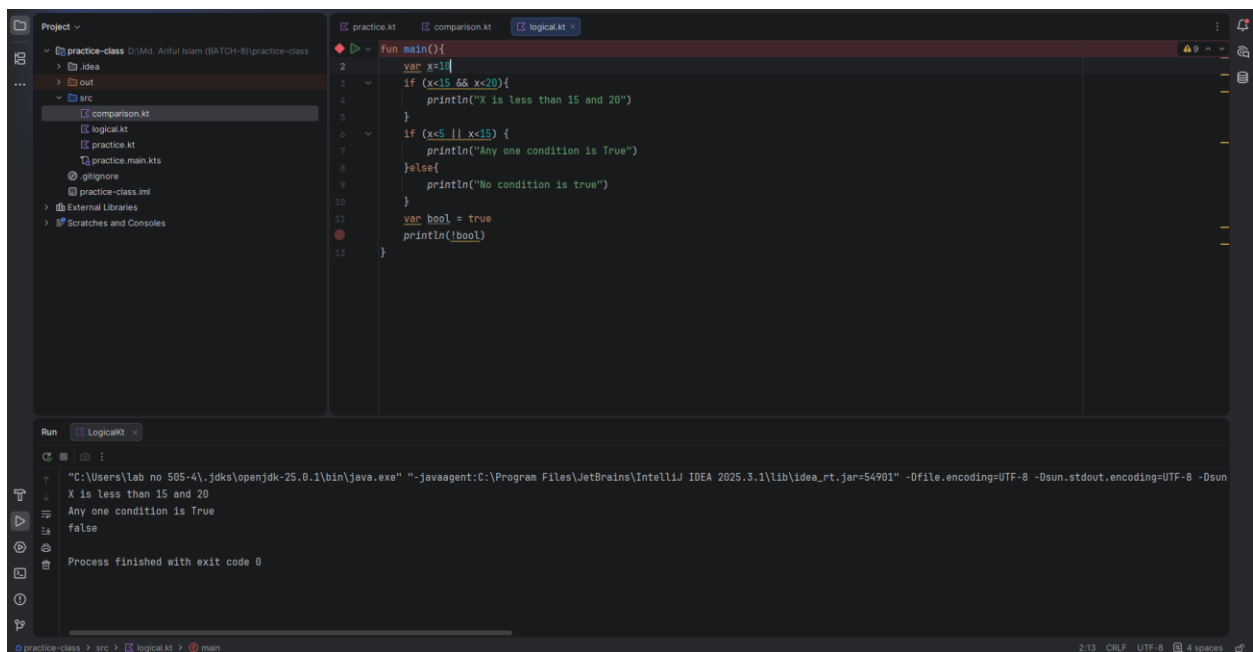
```
1 fun main(){
2     var x : Int = 10
3     var y : Int = 20
4     var z : Int=13
5
6     val sum = x+y
7     println("sum = $sum")
8     val sub = x - y
9     println("sub = $sub")
10    val mul = x*y
11    println("multiplication = $mul")
12    val div = y/x
13    println("div = $div")
14    val modulo = y%z
15    println("modulo = $modulo")
16
17    x = x++
18    println("x = $x")
19    y = y--
20    println("y = $y")
21 }
```

Run Practicekt

```
sum = 30
sub = -10
multiplication = 200
div = 2
modulo = 7
x = 10
y = 20

Process finished with exit code 0
```

Logical Operator



The screenshot shows the IntelliJ IDEA IDE with a project named 'practice-class'. The 'src' directory contains a file 'logical.kt'. The code in this file defines a 'main' function that uses logical operators to check conditions. It uses 'if' statements with '&&' and '||' operators to print messages based on the values of 'x' and 'y'. The Run window at the bottom shows the output of these logical operations.

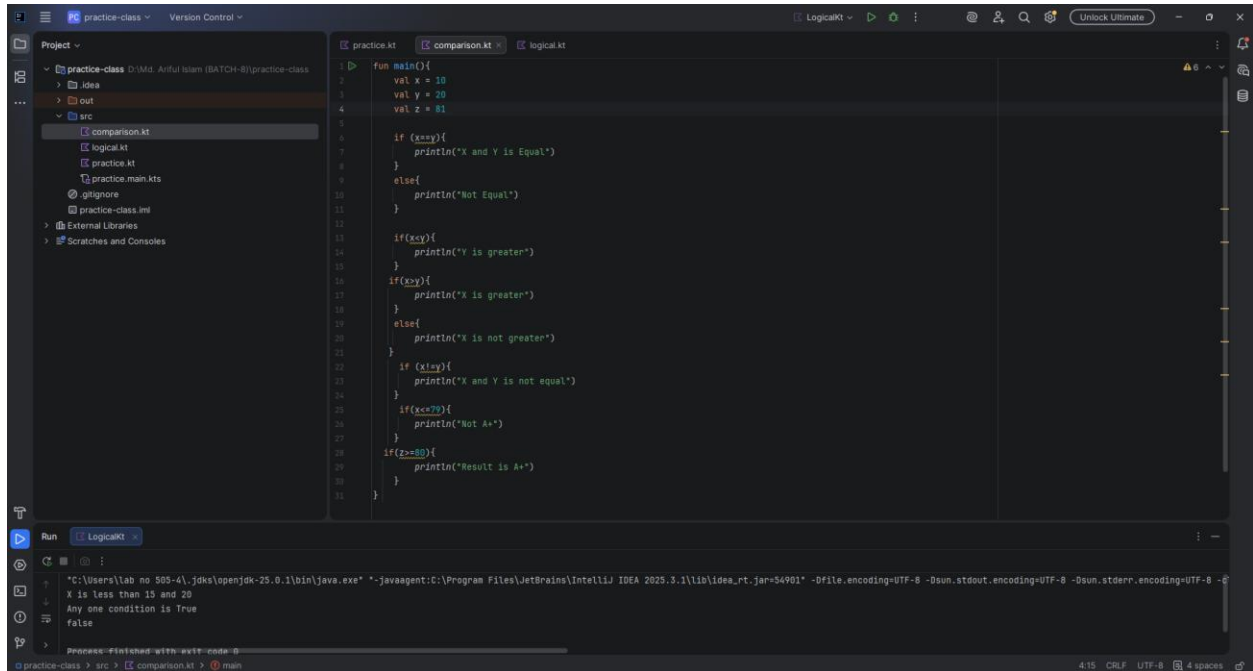
```
1 fun main(){
2     var x=14
3     if (x<15 && x<20){
4         println("x is less than 15 and 20")
5     }
6     if (x<5 || x<15) {
7         println("Any one condition is True")
8     }else{
9         println("No condition is true")
10    }
11    var bool = true
12    println(!bool)
13 }
```

Run Logicalkt

```
"C:\Users\lab no 505-4\jdk\openjdk-25.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.1\lib\idea_rt.jar=54901" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun
x is less than 15 and 20
Any one condition is True
false

Process finished with exit code 0
```

Comparison Operator



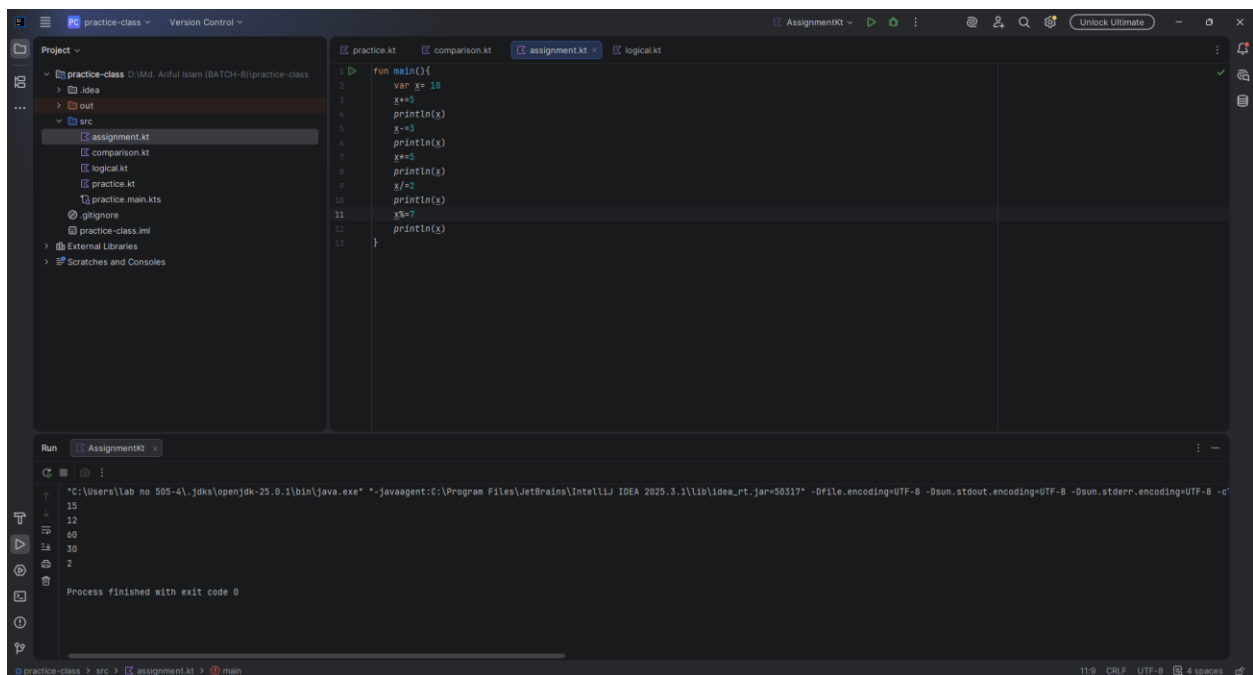
The screenshot shows the IntelliJ IDEA IDE with a project named 'practice-class'. The 'src' directory contains three files: 'comparison.kt', 'logical.kt', and 'practice.main.kts'. The 'comparison.kt' file is open, showing a Kotlin program that uses comparison operators to check the relationship between variables x, y, and z.

```
1 fun main() {
2     val x = 10
3     val y = 20
4     val z = 81
5
6     if (x==y) {
7         println("X and Y is Equal")
8     }
9     else {
10        println("Not Equal")
11    }
12
13    if (x<y) {
14        println("X is greater")
15    }
16    if (z>y) {
17        println("X is greater")
18    }
19    else {
20        println("X is not greater")
21    }
22
23    if (x!=y) {
24        println("X and Y is not equal")
25    }
26    if (x<=70) {
27        println("Not A+")
28    }
29    if (z>=80) {
30        println("Result is A+")
31    }
32 }
```

The Run console shows the output of the program:

```
*C:\Users\lab no 505-4\jdk\openjdk-25.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.1\lib\idea_rt.jar=54901" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -c
X is less than 15 and 20
Any one condition is True
false
Process finished with exit code 0
```

Assignment Operator



The screenshot shows the IntelliJ IDEA IDE with the same project 'practice-class'. The 'src' directory now includes 'assignment.kt' along with the previous files. The 'assignment.kt' file is open, showing a Kotlin program that uses assignment operators to calculate the value of x.

```
1 fun main() {
2     var x = 10
3     x+=5
4     println(x)
5     x-=5
6     println(x)
7     x*=5
8     println(x)
9     x/=2
10    println(x)
11    x%=7
12    println(x)
13 }
```

The Run console shows the output of the program:

```
*C:\Users\lab no 505-4\jdk\openjdk-25.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.1\lib\idea_rt.jar=50317" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -c
15
12
60
30
2
Process finished with exit code 0
```

